

Pneumonia Vaccine Navigator

Simplified, complete worked example

Module F • Participant guide

Synthetic teaching data

Instats • August 5, 2026

Among adults in a synthetic health-system cohort, what is the effect of pneumococcal vaccination at baseline, versus no vaccination, on 12-month pneumonia hospitalization?

A=1

vaccination

A=0

no vaccination

Y

12-month
hospitalization

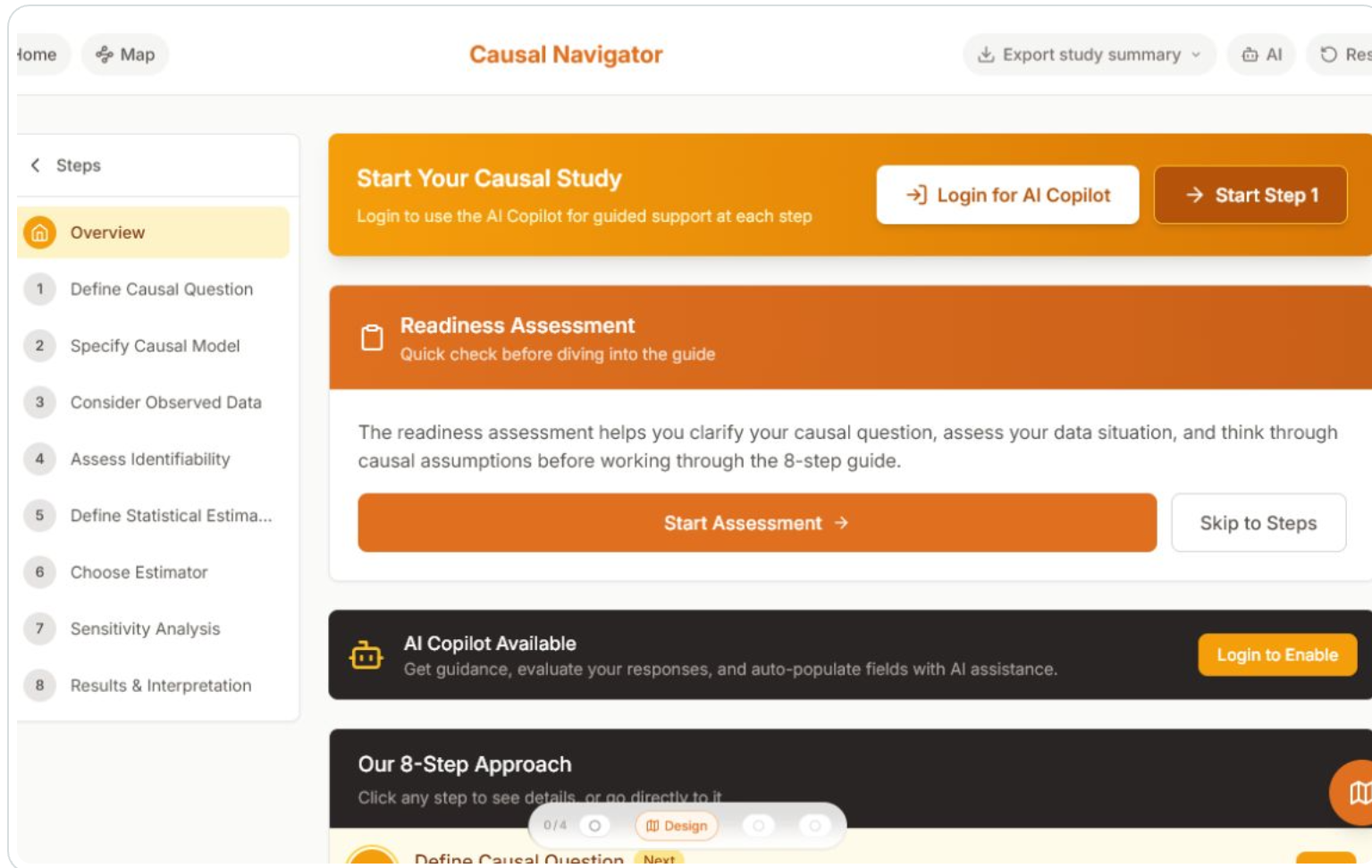
Target

average risk difference

Synthetic data only: the result teaches workflow, not clinical effectiveness.

The Navigator is an eight-step spine

Overview



1 Question

2 Causal model

3 Observed data

4 Identification

5 Estimand

6 Estimator

7 Sensitivity

8 Interpretation

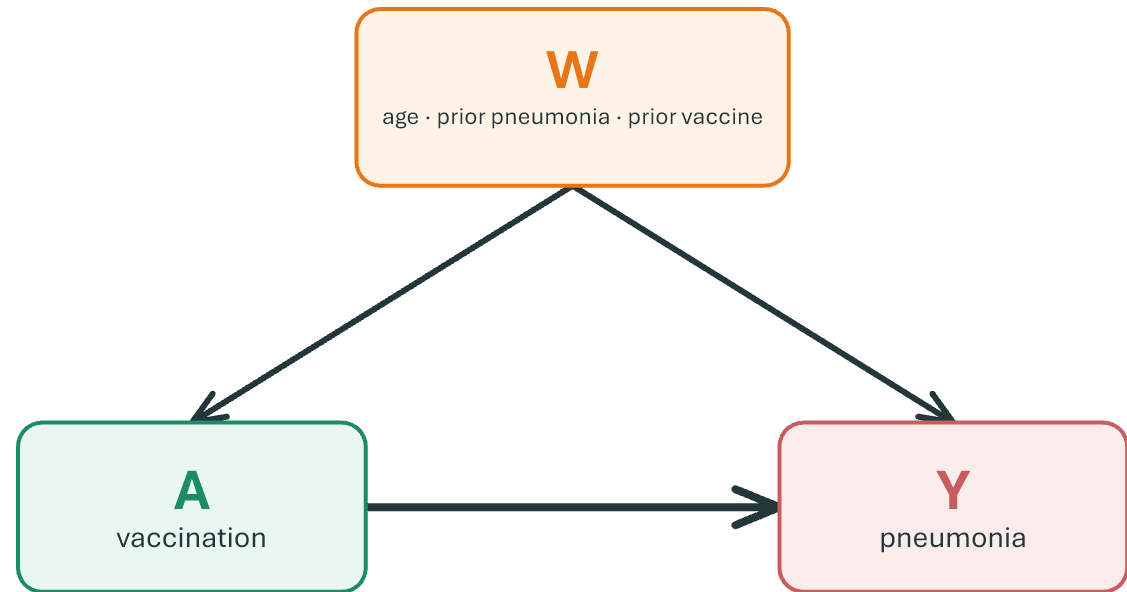
Each answer constrains the next.

Steps 1-2: protocol first, then the DAG

Question + model

Protocol

- **Population:** 5,000 simulated adults.
- **Treatment:** vaccination at baseline.
- **Outcome:** pneumonia hospitalization by 12 months.
- **Contrast:** average risk difference.



Adjustment set: age, prior pneumonia, and prior vaccination.

Steps 3-4: map the data, then audit identification

Data + assumptions

5,000

people

1,627

vaccinated

334

events

Point-treatment cohort · no missingness · complete 12-month outcomes

Identification assumptions

- **Exchangeability:** holds given W in the simulation.
- **Positivity:** propensities stay about 0.08-0.91.
- **Consistency:** one baseline treatment version.
- **No interference:** assumed.

In real EHR data, every one of these needs evidence.

CAUSAL TARGET

$$E[Y^1] - E[Y^0]$$

STATISTICAL ESTIMAND

$$E_x\{E(Y|A=1,W) - E(Y|A=0,W)\}$$

Primary estimator

TMLE

- $Q(A,W)$: outcome regression
- $g(W)$: treatment mechanism
- **Learners:** GLM, glmnet, ranger
- **CI:** efficient influence function

The estimator targets the quantity already chosen.

Step 7: plan how the conclusion could fail

Sensitivity

Unmeasured confounding

Could residual frailty or healthcare-seeking move the RD by +3 points?

Outcome error

Would imperfect pneumonia coding change direction or magnitude?

Selection

What if death, disenrollment, or EHR capture is informative?

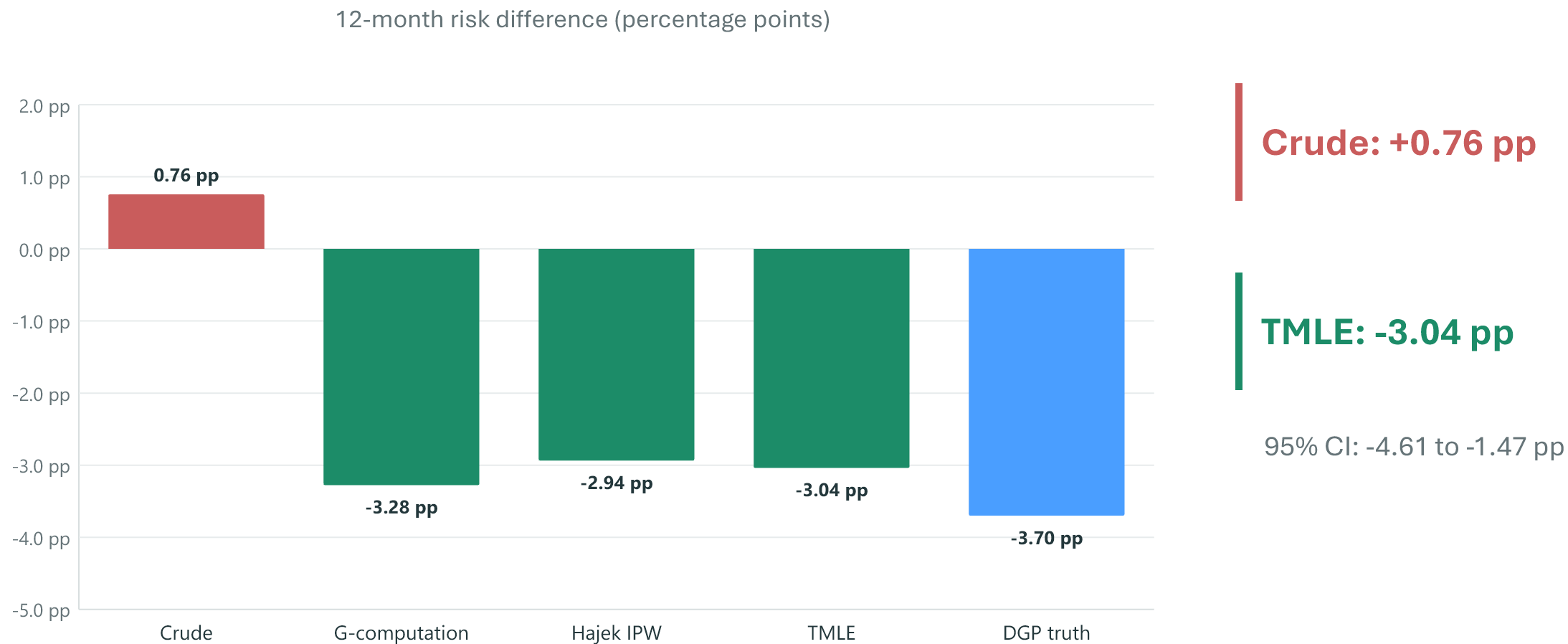
Estimation

Do alternative learners or propensity truncation change the result?

A confidence interval does not measure these causal gaps.

The sign reversal is the teaching result

Results



Synthetic teaching result - not clinical evidence

Supported

In this known simulation, adjustment reverses the crude association and recovers a protective direction close to the causal truth.

Still unknown

- Clinical vaccine effectiveness
- Safety
- Transportability to patients
- Real EHR confounding and measurement error
- Population transmission effects

The defensible conclusion is about the workflow, not the vaccine.

Choose one field from Steps 1-8.

- 1** What evidence makes the entry defensible?
- 2** What would make you revise it?
- 3** Which downstream decision changes if you revise it?

Use the completed roadmap as an answer key - not as a substitute for your argument.